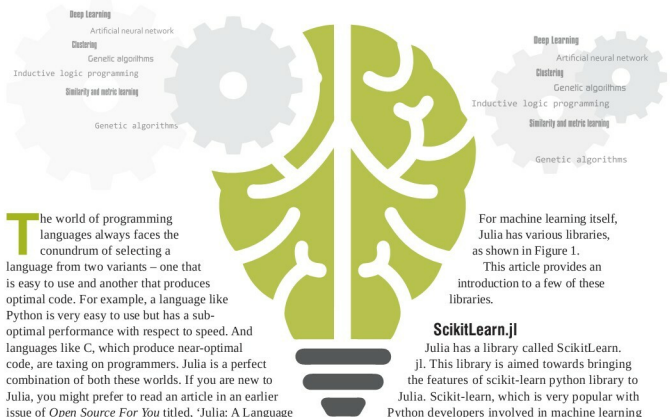


Machine Learning: The Julia Approach

Julia is a rising star in the world of programming languages, having inherited the best features of both Python and C. This article provides insights into the machine learning related features offered by Julia. As it is optimised for both speed and ease-of-use, developing machine learning based applications in Julia is fairly straightforward.



The world of programming languages always faces the conundrum of selecting a language from two variants – one that is easy to use and another that produces optimal code. For example, a language like Python is very easy to use but has a sub-optimal performance with respect to speed. And languages like C, which produce near-optimal code, are taxing on programmers. Julia is a perfect combination of both these worlds. If you are new to Julia, you might prefer to read an article in an earlier issue of *Open Source For You* titled, 'Julia: A Language that Walks like Python and Runs like C'. In addition to this, there are plenty of resources on the Web that enable a developer to become familiar with Julia (<http://docs.julialang.org/en/release-0.5/>).

The focus of this article is to introduce readers to machine learning and associated tasks in Julia. The core idea of machine learning is to enable machines to model the problem space in such a way that programs are capable of handling novel scenarios. For example, an artificial neural network for digit recognition should be able to recognise the variations of digits that were not present in the training dataset. In simple words, these networks are bio-inspired, similar to the manner in which biological creatures respond to scenarios, after successful training. Machine learning can be used across various domains such as text recognition, image recognition, classification, etc.

Similar to Python, Julia is also extensible. It has many libraries and packages focusing on specific domains.

For machine learning itself, Julia has various libraries, as shown in Figure 1.

This article provides an introduction to a few of these libraries.

ScikitLearn.jl

Julia has a library called ScikitLearn.jl. This library is aimed towards bringing the features of scikit-learn python library to Julia. Scikit-learn, which is very popular with Python developers involved in machine learning based projects and has implemented the *scikit-learn* algorithms. The major features of *ScikitLearn.jl* are listed below:

- It has around 150 models accessed with a uniform interface
 - Pipelines and FeatureUnions
 - Cross-validation
 - Hyperparameter tuning
 - DataFrames support
- Installation of *ScikitLearn.jl* may be done using the following command:

```
Pkg.add("ScikitLearn")
```

It has to be noted that the scikit-learn Python library is required to import the optional Python modules. *Pyplot.jl* is also utilised. As noted in the official documentation, *ScikitLearn.jl* is not associated with *scikit-learn.org*.